# Confidence Intervals & Hypothesis Testing

## Constructing a confidence interval

**1** A random sample of 50 students has a mean study time of $\bar{x} = 6.2$ hours per week, with sample standard deviation $s = 1.8$ hours. Construct a 95% confidence interval for the population mean study time, using a critical value of $z^* \approx 1.96$.

**2** Interpret the confidence interval from the previous problem in the context of the study time question, using correct statistical language.

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## Margin of error and sample size

**3** Explain two ways to decrease the margin of error of a confidence interval, and state a tradeoff involved with each.

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## The logic of hypothesis testing

**4** State the null hypothesis $H_0$ and alternative hypothesis $H_a$ for a test of whether a coin is biased toward heads (true proportion of heads $p$), and classify the test as one-tailed or two-tailed.

**5** Explain what a $p$-value represents, in the context of a hypothesis test.

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## Four-step hypothesis testing procedure

**6** A manufacturer claims their light bulbs last on average $\mu = 1000$ hours. A sample of 40 bulbs has mean lifetime $\bar{x} = 985$ hours with sample standard deviation $s = 45$ hours. State the hypotheses and compute the test statistic for testing whether the true mean lifetime is less than claimed.

**7** Using the test statistic $z \approx -2.11$ from the previous problem and a $p$-value of approximately $0.017$, state a conclusion at the $\alpha = 0.05$ significance level.

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## Type I and Type II errors, and power

**8** In the light bulb test above, describe what a Type I error would mean in context, and what a Type II error would mean in context.

**9** Explain what the power of a test measures, and state one way to increase it.

## Confidence intervals for a proportion

**10** A poll of 400 randomly selected voters finds that 220 support a candidate. Construct a 95% confidence interval for the true population proportion, using $z^* \approx 1.96$.

## Determining sample size for a target margin of error

**11** A researcher wants a margin of error of at most $0.03$ for a 95% confidence interval for a population proportion, using a conservative planning value of $p = 0.5$ (which maximizes required sample size). Find the minimum sample size needed, using $n = \left(\frac{z^*}{m}\right)^2 p(1-p)$ with $z^* = 1.96$.

## One-sample t-test for a mean

**12** A nutritionist claims a new granola bar has a mean of $200$ calories. A sample of 15 bars has mean $\bar{x} = 206$ calories with sample standard deviation $s = 9$ calories. Since the population standard deviation is unknown and $n$ is small, a $t$-test is used. Compute the test statistic $t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$.

**13** Explain what it means for a hypothesis test result to be "statistically significant," and why statistical significance does not always imply practical importance.

**14** A researcher wants to test whether a new fitness program changes the average resting heart rate of participants from the population baseline of $\mu = 72$ bpm. A sample of 36 participants after the program has mean resting heart rate $\bar{x} = 68.5$ bpm with sample standard deviation $s = 10.5$ bpm. State the hypotheses (choosing a two-sided alternative), compute the test statistic, and using a $p$-value of approximately $0.11$, state a conclusion at $\alpha = 0.05$.

**15** Explain the relationship between a confidence interval and a two-sided hypothesis test at the same confidence/significance level — specifically, describe how you could use a 95% confidence interval alone to decide whether to reject $H_0 : \mu = \mu_0$ at $\alpha = 0.05$, without separately computing a $p$-value.

**16** A city council is deciding whether to fund a new public transit line, and wants to survey residents to estimate the true proportion who would use it regularly. A preliminary small survey suggests around 35% support, but the council wants a formal survey with a margin of error no larger than 2.5% at 95% confidence. Using $n = \left(\frac{z^*}{m}\right)^2 p(1-p)$ with the preliminary estimate $p = 0.35$ and $z^* = 1.96$, find the required sample size, rounding appropriately.

17 A restaurant chain wants to test whether a new menu item's average preparation time is longer than the target of 8 minutes, since customers have complained about slow service. A random sample of 30 orders for the new item has a mean preparation time of $\bar{x} = 8.7$ minutes with sample standard deviation $s = 1.8$ minutes. State the hypotheses for this one-sided test, compute the test statistic, and using a $p$-value of approximately $0.019$, state a conclusion at $\alpha = 0.05$ in the context of the restaurant's concern.